No.	Status	Date
LNDE-RPT-96-014-DS-4-LA-104	Issued	1996-03-08

UT Thickness Data Sheet - 4-LA-104

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurements for line 4-LA-104, located within the ATU-100. The objective is to establish baseline wall thickness data for this specific line as part of the pre-service inspection program.

Grace Okafor, acting as the UT technician (SNT-TC-1A UT Level II) for Larkhall NDE Services Inc., performed the measurements and is responsible for the accuracy of the recorded data.

Technical Basis

All thickness measurements were conducted in accordance with the site NDE procedure QSCP-NDT-001 Rev 0. The inspection methodology follows the guidelines established in ASTM E797.

The data recorded herein supports the findings detailed in the baseline UT thickness survey LNDE-RPT-96-014, which covers 118 CMLs for the unit.

Component Specifications

The subject line, 4-LA-104, is an A106 Gr.B L/R exchanger lean side line. It has a nominal outer diameter of 114.3 mm and a nominal wall thickness of 6.02 mm.

The design parameters for this line are 2758 kPa(g) and 138 °C. The minimum allowable thickness (t-min) is 3.18 mm.

Inspection Results

Thickness readings were taken at designated points to verify the baseline condition of the pipe wall. As the ATU-100 is not yet in service, these readings represent the initial state of the asset.

All measured values are compared against the t-min of 3.18 mm to ensure the line meets the required thickness specifications prior to startup.

Thickness readings, measured 1996-03-08

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	—	—	7.12	—	—	—	2001-03-08	
6-SG-101-02	7.11	3.56	—	—	7.05	—	—	—	2001-03-08	
6-SG-101-03	7.11	3.56	—	—	7.08	—	—	—	2001-03-08	
6-SG-101-04	7.11	3.56	—	—	7.09	—	—	—	2001-03-08	
6-SG-101-05	7.11	3.56	—	—	7.14	—	—	—	2001-03-08	
6-SG-101-06	7.11	3.56	—	—	7.07	—	—	—	2001-03-08	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	—	—	7.14	—	—	—	2001-03-08	
6-SG-102-02	7.11	3.56	—	—	7.11	—	—	—	2001-03-08	
6-SG-102-03	7.11	3.56	—	—	7.04	—	—	—	2001-03-08	
6-SG-102-04	7.11	3.56	—	—	6.99	—	—	—	2001-03-08	
6-SG-102-05	7.11	3.56	—	—	7.13	—	—	—	2001-03-08	

Line 8-TG-101 | Treated gas | A106 Gr.B | t_nom 8.18 mm | t-min 4.06 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
8-TG-101-01	8.18	4.06	—	—	8.20	—	—	—	2006-03-08	
8-TG-101-02	8.18	4.06	—	—	8.20	—	—	—	2006-03-08	
8-TG-101-03	8.18	4.06	—	—	8.23	—	—	—	2006-03-08	
8-TG-101-04	8.18	4.06	—	—	8.10	—	—	—	2006-03-08	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
8-TG-101-05	8.18	4.06	—	—	8.22	—	—	—	2006-03-08	

Line 4-LA-101 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-101-01	6.02	3.18	—	—	5.99	—	—	—	2006-03-08	
4-LA-101-02	6.02	3.18	—	—	6.01	—	—	—	2006-03-08	
4-LA-101-03	6.02	3.18	—	—	5.96	—	—	—	2006-03-08	
4-LA-101-04	6.02	3.18	—	—	6.07	—	—	—	2006-03-08	
4-LA-101-05	6.02	3.18	—	—	6.01	—	—	—	2006-03-08	
4-LA-101-06	6.02	3.18	—	—	6.06	—	—	—	2006-03-08	
4-LA-101-07	6.02	3.18	—	—	5.99	—	—	—	2006-03-08	

Line 4-LA-102 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-102-01	6.02	3.18	—	—	6.02	—	—	—	2006-03-08	
4-LA-102-02	6.02	3.18	—	—	6.07	—	—	—	2006-03-08	
4-LA-102-03	6.02	3.18	—	—	6.07	—	—	—	2006-03-08	
4-LA-102-04	6.02	3.18	—	—	5.99	—	—	—	2006-03-08	
4-LA-102-05	6.02	3.18	—	—	6.00	—	—	—	2006-03-08	
4-LA-102-06	6.02	3.18	—	—	6.00	—	—	—	2006-03-08	
4-LA-102-07	6.02	3.18	—	—	6.00	—	—	—	2006-03-08	

Line 6-LA-103 | Lean amine header | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-LA-103-01	7.11	3.56	—	—	7.02	—	—	—	2006-03-08	
6-LA-103-02	7.11	3.56	—	—	7.19	—	—	—	2006-03-08	
6-LA-103-03	7.11	3.56	—	—	7.08	—	—	—	2006-03-08	
6-LA-103-04	7.11	3.56	—	—	7.17	—	—	—	2006-03-08	
6-LA-103-05	7.11	3.56	—	—	7.19	—	—	—	2006-03-08	
6-LA-103-06	7.11	3.56	—	—	7.16	—	—	—	2006-03-08	
6-LA-103-07	7.11	3.56	—	—	7.18	—	—	—	2006-03-08	
6-LA-103-08	7.11	3.56	—	—	7.04	—	—	—	2006-03-08	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	—	—	8.52	—	—	—	2006-03-08	
4-RA-101-02	8.56	3.81	—	—	8.54	—	—	—	2006-03-08	
4-RA-101-03	8.56	3.81	—	—	8.61	—	—	—	2006-03-08	
4-RA-101-04	8.56	3.81	—	—	8.63	—	—	—	2006-03-08	
4-RA-101-05	8.56	3.81	—	—	8.60	—	—	—	2006-03-08	
4-RA-101-07	8.56	3.81	—	—	8.60	—	—	—	2006-03-08	
4-RA-101-08	8.56	3.81	—	—	8.68	—	—	—	2006-03-08	
4-RA-101-09	8.56	3.81	—	—	8.57	—	—	—	2006-03-08	
4-RA-101-10	8.56	3.81	—	—	8.67	—	—	—	2006-03-08	
4-RA-101-11	8.56	3.81	—	—	8.68	—	—	—	2006-03-08	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	—	—	8.54	—	—	—	2006-03-08	
4-RA-102-02	8.56	3.81	—	—	8.57	—	—	—	2006-03-08	
4-RA-102-03	8.56	3.81	—	—	8.56	—	—	—	2006-03-08	
4-RA-102-04	8.56	3.81	—	—	8.54	—	—	—	2006-03-08	
4-RA-102-05	8.56	3.81	—	—	8.55	—	—	—	2006-03-08	
4-RA-102-06	8.56	3.81	—	—	8.51	—	—	—	2006-03-08	
4-RA-102-07	8.56	3.81	—	—	8.58	—	—	—	2006-03-08	
4-RA-102-08	8.56	3.81	—	—	8.49	—	—	—	2006-03-08	
4-RA-102-09	8.56	3.81	—	—	8.55	—	—	—	2006-03-08	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	—	—	11.00	—	—	—	2006-03-08	
6-RA-103-02	10.97	4.57	—	—	11.09	—	—	—	2006-03-08	
6-RA-103-03	10.97	4.57	—	—	10.99	—	—	—	2006-03-08	
6-RA-103-04	10.97	4.57	—	—	10.98	—	—	—	2006-03-08	
6-RA-103-05	10.97	4.57	—	—	10.95	—	—	—	2006-03-08	
6-RA-103-06	10.97	4.57	—	—	11.01	—	—	—	2006-03-08	
6-RA-103-07	10.97	4.57	—	—	10.99	—	—	—	2006-03-08	
6-RA-103-08	10.97	4.57	—	—	10.97	—	—	—	2006-03-08	
6-RA-103-09	10.97	4.57	—	—	11.01	—	—	—	2006-03-08	
6-RA-103-10	10.97	4.57	—	—	11.02	—	—	—	2006-03-08	

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	—	—	7.59	—	—	—	2001-03-08	
3-RV-101-02	7.62	3.56	—	—	7.57	—	—	—	2001-03-08	
3-RV-101-03	7.62	3.56	—	—	7.52	—	—	—	2001-03-08	
3-RV-101-04	7.62	3.56	—	—	7.59	—	—	—	2001-03-08	
3-RV-101-05	7.62	3.56	—	—	7.67	—	—	—	2001-03-08	
3-RV-101-06	7.62	3.56	—	—	7.59	—	—	—	2001-03-08	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	—	—	5.53	—	—	—	2001-03-08	
2-RV-102-02	5.54	2.79	—	—	5.59	—	—	—	2001-03-08	
2-RV-102-03	5.54	2.79	—	—	5.48	—	—	—	2001-03-08	
2-RV-102-04	5.54	2.79	—	—	5.58	—	—	—	2001-03-08	

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	—	—	5.48	—	—	—	2006-03-08	
3-RF-101-02	5.49	2.41	—	—	5.45	—	—	—	2006-03-08	
3-RF-101-03	5.49	2.41	—	—	5.45	—	—	—	2006-03-08	
3-RF-101-04	5.49	2.41	—	—	5.46	—	—	—	2006-03-08	
3-RF-101-05	5.49	2.41	—	—	5.45	—	—	—	2006-03-08	
3-RF-101-06	5.49	2.41	—	—	5.52	—	—	—	2006-03-08	
3-RF-101-07	5.49	2.41	—	—	5.42	—	—	—	2006-03-08	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	—	—	3.92	—	—	—	2006-03-08	
2-RF-102-02	3.91	2.16	—	—	3.88	—	—	—	2006-03-08	
2-RF-102-03	3.91	2.16	—	—	3.93	—	—	—	2006-03-08	
2-RF-102-04	3.91	2.16	—	—	3.87	—	—	—	2006-03-08	
2-RF-102-07	3.91	2.16	—	—	3.96	—	—	—	2006-03-08	

Line 4-LA-104 | L/R exchanger, lean side | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-104-01	6.02	3.18	—	—	6.07	—	—	—	2006-03-08	
4-LA-104-02	6.02	3.18	—	—	6.03	—	—	—	2006-03-08	
4-LA-104-03	6.02	3.18	—	—	5.96	—	—	—	2006-03-08	
4-LA-104-04	6.02	3.18	—	—	6.06	—	—	—	2006-03-08	
4-LA-104-05	6.02	3.18	—	—	6.04	—	—	—	2006-03-08	
4-LA-104-06	6.02	3.18	—	—	6.10	—	—	—	2006-03-08	
4-LA-104-07	6.02	3.18	—	—	6.01	—	—	—	2006-03-08	
4-LA-104-08	6.02	3.18	—	—	5.94	—	—	—	2006-03-08	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	—	—	8.51	—	—	—	2006-03-08	
4-RA-104-02	8.56	3.81	—	—	8.55	—	—	—	2006-03-08	
4-RA-104-03	8.56	3.81	—	—	8.54	—	—	—	2006-03-08	
4-RA-104-04	8.56	3.81	—	—	8.59	—	—	—	2006-03-08	
4-RA-104-05	8.56	3.81	—	—	8.60	—	—	—	2006-03-08	
4-RA-104-06	8.56	3.81	—	—	8.59	—	—	—	2006-03-08	
4-RA-104-07	8.56	3.81	—	—	8.59	—	—	—	2006-03-08	

Line 2-ST-101 | Reboiler steam | A106 Gr.B | t_nom 3.91 mm | t-min 2.29 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-ST-101-01	3.91	2.29	—	—	3.93	—	—	—	2006-03-08	
2-ST-101-02	3.91	2.29	—	—	3.90	—	—	—	2006-03-08	
2-ST-101-03	3.91	2.29	—	—	3.94	—	—	—	2006-03-08	
2-ST-101-04	3.91	2.29	—	—	3.82	—	—	—	2006-03-08	
2-ST-101-05	3.91	2.29	—	—	3.91	—	—	—	2006-03-08	

Line 3-CW-101 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-101-01	5.49	2.79	—	—	5.42	—	—	—	2006-03-08	
3-CW-101-02	5.49	2.79	—	—	5.41	—	—	—	2006-03-08	
3-CW-101-03	5.49	2.79	—	—	5.53	—	—	—	2006-03-08	
3-CW-101-04	5.49	2.79	—	—	5.46	—	—	—	2006-03-08	
3-CW-101-05	5.49	2.79	—	—	5.48	—	—	—	2006-03-08	

Line 3-CW-102 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-102-01	5.49	2.79	—	—	5.50	—	—	—	2006-03-08	
3-CW-102-02	5.49	2.79	—	—	5.51	—	—	—	2006-03-08	
3-CW-102-03	5.49	2.79	—	—	5.53	—	—	—	2006-03-08	
3-CW-102-04	5.49	2.79	—	—	5.51	—	—	—	2006-03-08	

Sign-off	Name	Function	Date	Signature
Prepared	Grace Okafor	UT technician, SNT-TC-1A UT Level II	1996-03-08	_____